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SHORT-PAPER

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Abstract

As emerging technologies such as AI systems and social robots become embedded in children's daily lives, it is vital to consider how children themselves imagine robots. Advances in social robotics have generated new opportunities for assessing child wellbeing, but much of this work remains framed by adult researcher-centered views of what counts as beneficial. Drawing on the distinction between adult child perspectives and children's own perspectives from childhood studies, we report a speculative participatory workshop with a child public involvement panel that explores future social robots for supporting children's wellbeing. Children took part as co-researchers rather than research subjects, and their discussions highlight insights including everyday care, emotional companionship, and transparency about robot identity. We use these insights to inform future work on child-robot interaction and child wellbeing, and to connect children's perspectives to debates around ethics and AI policy.

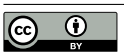
CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics** → **Children**.

Keywords

Human-Robot Interaction, Child-Robot Interaction, Design fiction, Participatory Design, Speculative Design, Ethics, Child Wellbeing

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1 Introduction

As artificial intelligence (AI) and robotic technologies become increasingly integrated into daily life, children growing up in this era will live alongside intelligent systems in ways that are difficult to predict. One recent study in *Nature* proposes "science-fiction-science" (sci-fi-sci) as a way to anticipate the social and behavioral impact of future technologies [25]. In sci-fi-sci, present-day participants are immersed in controlled simulations of speculative technologies that do not yet exist, in order to measure behavioral, psychological and social effects. This idea resonates with speculative design work in Human-Robot Interaction (HRI), where much of the field is implicitly speculative [31]. It also raises a question about who gets to imagine these futures: science fiction writers and researchers, or also the people who will live with and use these technologies?

Prior child-robot interaction work has shown that social robots can support children in education [5] and healthcare domains [21, 24]. Recent research also points to the potential use of social robots within the area of child wellbeing, both for intervention and for assessment [1, 16]. However, most of these studies utilize predefined scenarios designed by adults, with limited involvement of children in the early stages of imagining or shaping what the robots should be like, even though children are the ones most directly affected by such technologies [2]. This is particularly important when robots are linked to children's wellbeing, which is not only a clinical or adult-defined construct but also something children experience and interpret in their own way.

As UNICEF emphasizes in their Policy Guidance on AI for children [28], a key element of realizing children's participation rights is to undertake research with children, not only treating them as subjects. Hart's ladder of participation describes different levels of children's involvement, from manipulation to child initiated, shared decision-making with adults [15]. It can be used as a guide when considering how to include children and young people in co-design and decision processes. Including children's subjective perspectives is crucial, since previous research suggests that ignoring children's own views on their wellbeing leads to bias: domains that matter to children risk being underrepresented or neglected [13]. Researchers like us may aim to design for children, but Sommer et al. [27] distinguish between child perspectives (adults' interpretations of children's lives) and children's perspectives (children's own views and experiences). Our work aims to move towards children's

perspectives by focusing on how children themselves perceive their lives, rather than only how adults think about them.

Speculative methods in HRI have clear potential for participatory co-creation with people outside the robotics research community [31]. However, adapting these methods for children is challenging, particularly when exploring abstract concepts like child wellbeing, a domain where as noted above children's own perspectives are often overlooked. The novelty of this work lies in integrating participatory design into speculative research to amplify children's voices, using drawing not just as a playful activity, but as an epistemological tool to elicit their ideas. In this project, we treat children's imaginaries as part of the mutual shaping of society and technology [33], and we aim to empower children to take an active role rather than only being recipients of social robots. The specific goal of this workshop as an initial step is to employ this visual-based speculative method with a child public involvement panel to investigate how children conceptualize social robots for wellbeing. In this context, wellbeing is understood broadly as what makes life feel good, safe, and supported from their point of view. We aim to identify the roles, relationships, and expectations they attach to "good" robots.

2 Related Works

Participatory approaches offer ways to treat children as design partners rather than just users [12]. In HRI, such approaches often involve creative activities with young people, such as drawing, low-fidelity prototyping and role play [2, 6, 23]. For example, Mott et al. [22] combined storytelling to introduce abstract ethics-related topics with subsequent sketching and prototyping, and invited children to conduct video interviews to support power sharing. Co-design workshops can also bring children into conversations about children's rights and AI. Levinson et al. [17] used UNICEF's framework on children's rights and AI to structure workshops where children discussed fairness, inclusion and privacy in child-robot interaction, and their priorities informed recommendations for social robot design and AI policy.

Speculative design and design fiction have been adopted in HRI to explore desirable future sociotechnical systems [31]. Researchers have employed diverse narrative methods, such as speculative role-play [7] and fictional documentaries [11], to examine robots in complex social scenarios. With children, speculative methods have been used to elicit imaginaries such as hopes and fears about robot morphology [10]. When combined with participatory approaches, design fiction can also support child empowerment. Ventä-Olkkonen et al. [30] demonstrated how critical design fiction can empower children to reimagine sensitive issues like school bullying prevention, benefiting both children and society.

While prior work has used participatory and speculative methods to elicit children's values and imaginaries, it is less reported how such contributions are carried forward as advisory input to shape concrete research materials in HRI, particularly for abstract domains such as child wellbeing. Building on design research in HRI showing how fictional materials can help bypass technical limitations and support ethical exploration [18], we use drawing-based speculation with a child public involvement panel to translate children's imaginaries into vignette drafts that can later prompt reflection among adults, while centering children's perspectives rather than adult interpretations.

3 Methodology

We conducted a participatory workshop to initiate the co-creation of speculative robot futures for child wellbeing. The workshop was carried out with an established child panel operating within a Patient and Public Involvement (PPI) framework, that routinely meets to advise on research projects. Children participated as project advisors rather than study participants, and their contributions will be used to shape the development of future research materials (vignette drafts) rather than to generate data to answer predefined research questions. Accordingly, we report the workshop as advisory input for vignette development, not as an experimental dataset. The session also served as an initial familiarization with the robots to prepare the children for their ongoing advisory role in the project.

The workshop involved seven children (aged 4-11; 6 girls, 1 boy) from this child panel. They already knew each other and three of the researchers from previous meetings, which helped create a familiar and relaxed atmosphere. Although they had no prior experience interacting with social robots, their active role as PPI group members enabled them to generate complex ideas and effectively engage in speculative discussions about future technologies. The workshop was conducted in Swedish, the native language of the participating children, to ensure comfort and natural expression of their ideas. It took place in the Social Robotics Lab at Uppsala University after school and lasted around 90 minutes, to help maintain their attention.

Following recommendations for child-robot co-design [2], we assembled a multidisciplinary team of five researchers. HRI researchers managed the robotic systems and explained technical issues, while researchers from the Child Health and Parenting (CHAP) research group experienced in participatory research facilitated the sessions, supported the child panel and took notes of the discussion and developing ideas. As power dynamics are important in child involvement [9], and inspired by feminist HRI [32], we aimed to balance power relations between children, robots, and adults. To ensure the children's ideas reflected their own voices, we introduced as few new adults as possible. Guardians (present for the youngest children) and researchers acted primarily as logistical supporters. During drawing activities, adults engaged in active listening and used non-directive open-ended questions solely to clarify children's intended meanings, strictly avoiding any suggestions that might influence the content. This allowed children to steer the conversation and exercise agency. Guardians were asked to remain in the background unless invited by their child. To support the later drafting of study material (vignettes), we retained the drawings with the children's permission and took notes throughout the workshop capturing children's verbal explanations, so that later summaries were grounded in the children's intended meanings rather than interpretations of the drawings alone.

3.1 Ethical Considerations

The activity described in this paper constitutes public involvement in research in which the children were involved solely as co-researchers (public contributors), not as research participants.

Thus, an ethical review under Swedish law was not applicable. Instead, we adhered to ethical best practices for involving children as public contributors.

Both children and their guardians had received information about the purpose of the child panel and the children's role, and participation was voluntary with guardians' agreement. Following the panel's procedures, they received a specific invitation to join this research project on social robotics. Consent was treated as situational, so before the workshop children were again informed about the activity and asked if they wished to participate. Similarly, they could decide whether and how their work, including drawings, would be used. In this paper, we present one of the drawings as an example of the work. The child co-researcher who produced the drawing has agreed to this, and chosen to be acknowledged with her full name.

3.2 Workshop Procedure

The workshop was divided into two main phases: an initial demonstration phase and a subsequent speculative drawing activity.

3.2.1 Phase 1: Demonstration and familiarization. Given the children's lack of prior experience with social robots, this phase focused on familiarization. We introduced two contrasting platforms: Furhat, a robotic head with a more realistic face and voice, and NAO, a small humanoid robot with a clearly robotic appearance. During the demonstration, children were invited to talk with the Furhat robot about their favourite robots and their hobbies, and were given the option to shake hands with NAO as a warm-up and ice-breaker [1, 4]. This phase was intended to introduce basic robot functionality, build familiarity, reduce novelty effects, and provide an initial conceptual grounding before engaging in open-ended speculation, rather than basing their ideas only on movie characters and popular media [3, 19].

Another intention of the demonstration was to let children experience technological limitations and possible failures. Considering children's strong imagination, they may not spontaneously infer a robot's limitations [29]. Therefore we showed and discussed simple failure cases, such as non-response or speech-recognition errors, to emphasize that robots are not always reliable and can also fail. This was intended to support critical reflection, and during the subsequent drawing activity we did not further constrain or challenge children's imagined scenarios. After the demonstrations, we briefly reminded the children of the overall project theme (robots for children's wellbeing) and the purpose of the children's involvement in the project. To help them grasp this abstract concept, we provided everyday examples, such as feeling good at school or being happy in daily life. This was followed by a short break with refreshments to give the children time to rest and process the information.

3.2.2 Phase 2: Speculative Drawing. The second half of the workshop focused on the speculative drawing session, with the purpose to start generating ideas for vignettes to be used in later focus group discussions with adult research participants. Drawing was chosen because it is a playful, minimally scripted activity that aligns with children's central modes of communication, and sketching is widely used as a core design activity in child-robot design [2]. We utilize drawing as an epistemological tool to elicit ideas that children cannot yet fully express through writing [14]. Practically, this method was also well-liked by most of the children in the child panel, and accessible for those who are not fully able to write independently. However, the children were given freedom to write instead, or in addition to drawing. To frame the purpose of the activity and of the overall research project, we used the following narrative prompt: "*Imagine you are creating a story or a film that takes place in the year 3000, and there is a child of the same age as you does not feel good. How would a good robot notice that, and what could it do to help?*". The children then started to draw individually on A3 paper. The adult researchers stayed nearby, talked with the children about their drawings to support expression, and helped to write down notes capturing the children's explanations and ideas that were not visible in the drawings alone.

We concluded the workshop after the children finished their drawings. This initial session focused on individual speculative creation to avoid overloading children and to give them time to form their own ideas before engaging in group discussion in a future workshop. After this first workshop, the child health researchers leading the child panel wrote a project plan for the children, summarising the purpose of the robot project and the children's involvement, using the example of Alfred Nobel (well-known to Swedish children as it is covered yearly in schools) to explain the concept of unintended harm with research and the idea of sci-fi-sci. We will continue vignette development in a follow-up workshop, building on this initial session once children have had time to form a mental model of this child–robot interaction use case.

In line with the PPI child panel's advisory role, we did not conduct a formal coding procedure or inter-rater reliability assessment. Instead, the drawings and accompanying notes of children's explanations were used to compile a descriptive summary that will inform the drafting of vignettes.

4 The Children's Initial Drafts of Robot Futures

After the workshop, one adult researcher summarised these ideas to support vignette development. The summary and initial vignette suggestions will be brought back to the child panel for continued development. Because children produced different numbers of drawings, we report counts at the child level based on what was expressed in the drawings and their explanations. Most ideas focused on everyday assistance and basic needs, with cooking robots being the most frequent example (3 of 7 children). Other everyday-assistance ideas included housework, support at school (e.g., a teacher), or building houses. Two children imagined pet-like robots, but only one linked the pet to emotions, which was also the only explicit reference to emotions. In line with our broad framing of wellbeing, we treated these forms of everyday care and practical support as wellbeing-related, which suggests an emphasis on a stable everyday life where children are taken good care of and their basic needs are fulfilled.

In addition, we highlight two illustrative vignettes based on selected drawings and discussion notes, capturing design-relevant considerations beyond everyday functionality that emerged both on paper and in conversation.

4.1 Illustrative Vignettes

4.1.1 Robot as Caring Pet. Two of the drawings showed future robots as pet-like companions, for example for people allergic to animals. The vignette in Figure 1 shows a robot cat that helps children express their emotions. In this drawing, the robot cat asks the child "How are you?" and, after receiving a positive answer, shows happiness with hearts in its eyes. The child also wanted the robot cat to change color depending on their mood. The idea was inspired by the child's own experience: they would like to have a cat, but their parent is allergic, so they cannot. By this idea, the child panel brought attention to emotional companionship as an important aspect in vignettes on future robots for child wellbeing.



Figure 1: Robot as Caring Pet by Edith Reinhammar.

4.1.2 Honesty about Being a Robot. Instead of focusing on a specific context, another child said they would like the robot to be honest about being a robot. They felt it was easier to talk about their emotions and feelings to a robot than to a human stranger. They preferred a humanoid robot over Furhat, which only has a head, but at the same time the robot should have a note on its shirt showing that it is a robot. Informed by this discussion, our vignettes will include robots that are clearly non-human, as honesty about the robot's identity appeared important for feeling comfortable when talking to it.

5 Discussion and Future Works

The purpose of this workshop was to begin a collaboration with children as advisors in developing study material, consisting of vignettes depicting social robots in future scenarios, which will later be used in focus group discussions with adult participants. Through the workshop, we gained insight into what wellbeing could mean for children, which broadened our adult perspective. Although we initially expected a stronger focus on mental wellbeing, the drawings mostly pointed to basic needs, which can be linked to the lower levels of Maslow's hierarchy of needs [20], such as physiological and safety needs. This observation resonates with Levinson et al. [17], who found that children prioritize physical safety among all rights under UNICEF's framework on children's rights and AI. At the same time, there is some ambiguity as to whether some children set aside the wellbeing framing and simply treated the task as imagining a generally helpful robot in their lives. Overall, the workshop reminded us to adopt a broad view of child wellbeing that includes basic needs and everyday support, and our continued research will take these aspects seriously. As this was an advisory activity with a child panel based in Sweden, we do not make claims about representativeness or generalisability. Future work can broaden these perspectives in other settings.

Another important point is about transparency, both in identity and in interaction. In discussing what our vignettes should look like, one child wanted the robot to look somewhat human-like but also to wear a clear note saying that it is a robot, and another expected regular feedback from the robot. These small examples resonate with broader discussions in AI ethics and policy about transparency and children's rights in AI systems [28], which stress that AI and robots should not mislead users about their nature. In our continued work, we will include examples of robots that are emotionally approachable but clearly non-human and honest about their identity and capabilities, which aligns with previous findings of Calvo-Barajas et al. [8].

Participatory methods have also been critiqued for still relying on adult-designed structures, which can limit children's agency. However, adult involvement does not have to be understood as purely negative, it can also support and scaffold children's participation when it is used reflexively and with care [26]. In our setting, adult facilitation supported participation but also inevitably shaped the interaction to some extent, so some influence cannot be fully avoided.

In the short term, our future work is to continue the discussion within the panel through a follow-up workshop that revisits and extends the children's speculations. In the longer term, our broader project will continue to explore participatory approaches for children to take part in shaping social robots for their own wellbeing, engaging with wider discussions and critique in this field and centering children's perspectives in these debates. Taken together, this work is a small step toward bringing children's own imaginaries into conversations about future social robots for wellbeing.

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